

Emilio Urbina Arredondo

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PROFESSIONAL SUMMARY

Data Science and Mathematics student with a strong foundation in statistical modeling, machine learning, and mathematical optimization. Experienced in translating complex, real-world problems into structured, scalable analytical solutions using Python, SQL, and R, with a focus on interpretability, rigorous validation, and systems-level thinking.

EDUCATION

Tecnológico de Monterrey (ITESM)
B.S. in Data Science and Mathematics

Monterrey, Mexico
Aug. 2024 – June 2028

PROJECTS

BioLit Copilot | *Python, Pydantic, PubMed API, MeSH, Astro/TypeScript*

- Designed and built a biomedical literature pipeline: retrieves papers from PubMed, extracts and links chemical/disease entities to MeSH, clusters and ranks findings by relevance to the query, and generates a deterministic, source-attributed answer
- Ran the project measurement-first: an LLM extractor lost to a deterministic baseline (F1 0.31 vs 0.62) among six similar rejected mechanisms logged across 23 architecture decision records, for \$2.48 in total paid API spend
- Shipped a public evidence-viewer site verifying every quoted excerpt against its exact source licence and attribution, after the build itself caught a licensing bug and triggered a corpus-wide audit

Interpretable Student Attrition Modeling | *Python, Regression, Statistical Inference*

- Built an end-to-end statistical modeling framework on 121,000+ institutional observations, designing linear and logistic regression models with a focus on interpretability over pure predictive optimization
- Integrated bias assessment and ethical considerations into the analysis, delivering a technical report and executive-level presentation with actionable recommendations

Adaptive Intelligent Traffic Control | *Python, MDPs, Reinforcement Learning*

- Developed an adaptive traffic signal control agent using Markov Decision Process modeling, defining state space, action space, and reward structure to minimize congestion under stochastic arrivals
- Implemented and benchmarked fixed-cycle, heuristic, and Value Iteration-optimal policies, evaluating average delay, throughput, and queue stability in a rigorous experimental framework

Heterogeneous Vehicle Routing Optimization | *GAMS/CPLEX, MILP, Python*

- Formulated a Mixed-Integer Linear Program (Priority-based HVRPTW) for an NGO's heterogeneous fleet, modeling a social-priority objective to balance cost minimization against equitable coverage of vulnerable communities
- Solved the exact model to proven optimality in GAMS/CPLEX and built a Clarke-Wright + 2-opt heuristic in Python as a solver-free benchmark, quantifying a 15% optimality gap

Digital Commerce Performance Dashboard | *Python, Streamlit, Power BI, ML*

- Built an end-to-end analytics pipeline over a 12,000+ order B2B ecommerce dataset, engineering KPIs (AOV, conversion rate, cart abandonment, RFM features) across 12 structured tables
- Trained and compared revenue forecasting, churn scoring, and demand prediction models, and delivered a five-page Streamlit + Plotly dashboard alongside a Power BI report with DAX measures and a star-schema data model

TECHNICAL SKILLS

Programming & Data: Python (pandas, NumPy, scikit-learn), SQL (advanced queries, aggregations), R

Statistical & Analytical Methods: Statistical Inference, Hypothesis Testing, Regression Modeling, Exploratory Data Analysis

Machine Learning: Supervised Learning, Reinforcement Learning (Q-Learning, MDP-based Control), Model Evaluation & Validation, Feature Engineering

Optimization & Decision Models: Linear Optimization (foundations), Mixed-Integer Linear Programming (MILP), Markov Decision Processes

Tools & Platforms: Git, GitHub, Jupyter Notebook, Power BI

CERTIFICATES & LANGUAGES

Certificates: IBM AI Developer (IBM, Feb. 2025); International Business Experience (Esade, Jul. 2023); Conscious Leadership (Conscious Business Center, Nov. 2025)

Languages: Spanish (Native), English (Professional Working), French (Intermediate), German (Basic)